

SIMATS ENGINEERING ASSESSMENT TOOL 1

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COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

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Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every

window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Short Answer Test

Q1. Given

- File Size = 150
- Bandwidth = 50Mbps
- Segment size = 1500 bytes
- Frame overhead = 40 bytes

A company needs to transfer a design file from a client computer to remote server over a network, to calculate the total number of segments into frames transmitted, the total transmission overhead introduced by the data link layer, and estimate the total transmission time with given data,

i) Number of segments

$$150 \text{ MB} = 150 \times 1,000,000 = 150,000,000 \text{ bytes}$$

$$\therefore \text{number of segments} \Rightarrow \frac{150,000,000}{1500} = 100,000 \text{ segments}$$

ii) Number of frames

One segment is placed in one frame

$$\therefore \text{frames} = 100,000$$

iii) Total Data Link layer overhead
overhead per frame = 40 bytes

$$\text{Total overhead} = 100,000 \times 40 = 4,000,000 \text{ bytes}$$

$\Rightarrow$ Therefore, the total data link layer overhead is 4 MB.

ii) Total data transmitted

Original file = 150 MB

Overhead = 4 MB

Total data transmitted is equal to 154 MB.

v) Transmission time

Total data = 154 MB

$$154 \times 8 = 1232 \text{ Mbits}$$

Bandwidth = 50 Mbps

$$\therefore \text{Time} = \frac{\text{Data}}{\text{Bandwidth}}$$

$$= \frac{1232}{50}$$

The transmission time = 24.64 seconds.

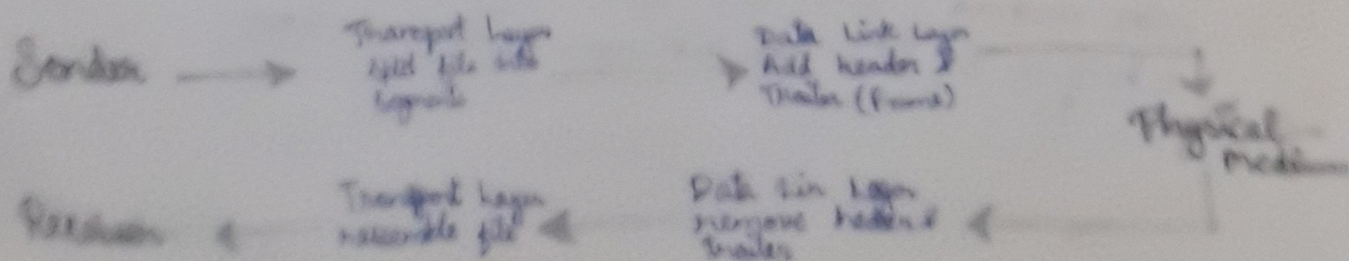
How Transport Layer ensures reliable communication.

The transport and data link layers are reliable communication as,
Transport Layer (Layer 4)

Transport layer is responsible for reliable end-to-end communication between the sender and receiver. As a layer 4, it divides large file into small segments, assigns sequence numbers, performs flow control, detects errors, and retransmits lost segments if necessary. It uses TCP and UDP as protocols.

Data Link Layer (Layer 2)

Data link layer transfers data between two directly connected devices. It converts segments into frames by adding a header and trailer, which contains addressing and error-checking information. It detects transmission errors using CRC and ensures reliable frame delivery.



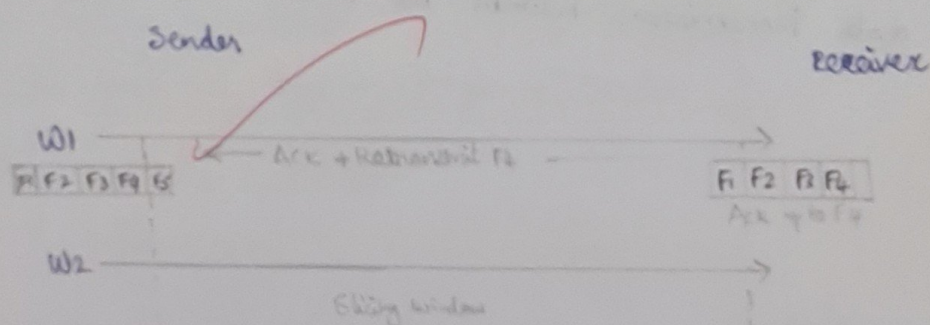
The transport layer ensures reliable end to end communication by segmenting, sequencing, and retransmitting lost data. The data link layer ensures reliable node to node communication by frame data, adding error detecting information and correcting transmission errors. Together, they provide efficient and reliable data transmission across the network.

2. Given:

- Sliding window size = 6 frame
- Each frame = 1024 bytes
- Total frames = 48
- One frame is lost in every window.

We know that, the sliding window protocol is a flow control technique used by the transport layer. It allows the sender to transmit multiple frames before receiving an acknowledgment. This improves communication speed and network efficiency.

If any frame is lost or corrupted, the receiver requests retransmission, ensuring reliable communication.



So to calculate total number of transmission window, retransmissions, user data transmitted;

$$\text{Window} = \frac{48}{6} = 8$$

∴ The number of transmission window in total is 8.

• Number of retransmissions

Number of windows = 8
since one frame is lost in each window
Therefore,

retransmissions = 8.

Total user data transmitted:

$$\Rightarrow 48 \times 1024 = 49,152 \text{ bytes.}$$

∴ The total user data transmitted is 49MB.

Flow Control Improves Communication as following:

- Prevents the sender from sending data too fast.
- Avoid receiver buffer overflow.
- Reduces packets loss.
- Improves network efficiency.
- Ensures reliable communication.
- Maintains smooth data transfer.

The sliding window protocol improves communication by allowing multiple frames to be transmitted before receiving any acknowledgments. Flow control prevents congestion, avoids data loss, and ensures efficient and reliable data transmission between the sender and receiver.

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header for each fragment = 20 bytes

The Network layer (Layer 3) is responsible for transporting packets from the source to the destination across different networks. It performs logical addressing (IP addressing), routing, fragmentation, and reassembly of packets.

To calculate the number of fragments created and total header overhead introduced, with given data,

→ Data in each fragment:

- MTU = 1500 bytes
- IP Header = 20 bytes
- Data in each fragment = $1500 - 20 = 1480$ bytes

→ Number of fragments

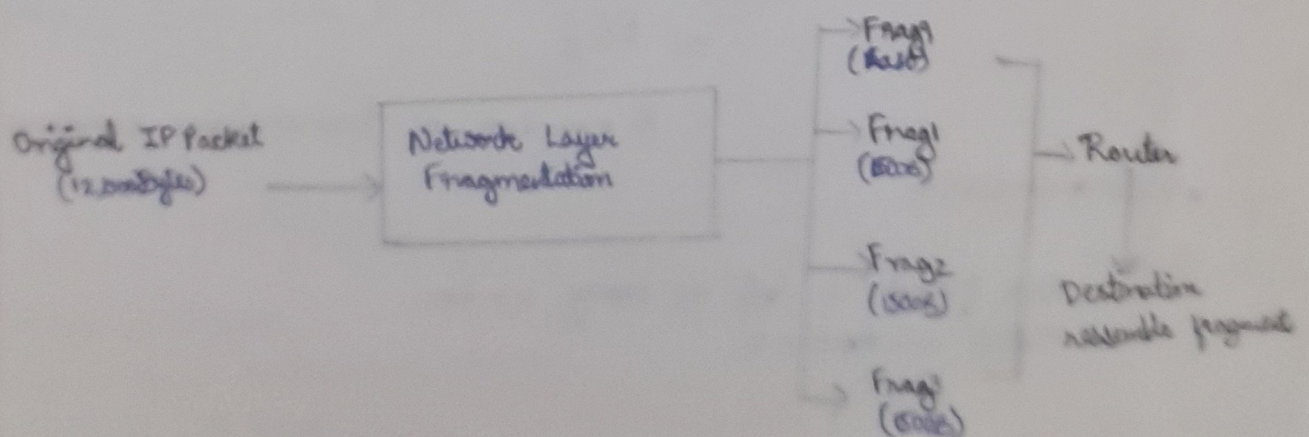
$$\text{Number of fragments} = 12,000 \div 1480 = 8.1$$

So, 9 fragments are required.

→ Total header overhead

Each fragment has a 20 byte IP header.

$$\begin{aligned}\text{Total overhead} &= 9 \times 20 \\ &= 180 \text{ bytes}\end{aligned}$$



How the Network Layer ensures successful delivery

- Assigns logical IP addresses.
- Selects the best path using routing.
- Fragments large packets to match the MTU.
- Adds fragment information for correct reassembly.
- Reassembles all fragments at the destination.
- Ensures reliable delivery across different network.

The Network Layer ensures successful communication by routing packets, fragmenting large packets to fit the MTU, and reassembling them at the destination. This allows data to travel efficiently and reliably across different networks.

4. Given

• Sliding window size = 6 frames

• Each frame = 1024 bytes

• Total frames = 48

• One frame is lost in every window

To find the total number of retransmissions:

$$\Rightarrow \text{Number of windows} = \text{Total frames} \div \text{Window size}$$

$$= 48 / 6$$

$$= 8 \text{ number of transmission windows.}$$

• Retransmissions

Given:

• One frame is lost in every window

• Total window = 8

Retransmissions = $2 \times 1 = 2$ frames

Total retransmissions = 2 frames

In total user data

• Each frame = 1024 bytes

• Total data = 48×1024

= 49,152 bytes

Advantages of flow control

- Prevents receiver buffer overflow
- Reduces packets loss
- Improves network performance
- Ensures reliable communication
- Maintains smooth data transmission.

Theory.

The sliding window protocol improves network efficiency by allowing multiple frames to be sent before receiving acknowledgments. Flow control prevents congestion, avoids data loss, and ensures reliable communication between the sender and receiver.

Conclusion,

Flow control is a transport layer mechanism that controls the speed of data transmission between the sender and receiver.

Flow control improves communication efficiency by matching the sender's transmission speed with the receiver's processing speed, resulting in faster, reliable, and smooth data transfer.

600MB Video File

0%	60	120	180	240	300
C ₁	C ₂	C ₃	C ₄	C ₅	C ₆

Failed at 200MB

Restart from checkpoint C₆ (360MB)

Retransmit only 200MB

Session

- Video file size = 600MB
- Checkpoint after every 60MB
- Failure occurs after 390MB transferred.

The session layer (no) establishes, manages, and terminates communication sessions between devices. It inserts synchronization checkpoints into data transfer.

• Number of checkpoints before failure:

$$\begin{aligned} \rightarrow \text{Checkpoints} &= \text{Data transferred before failure} \div \text{checkpoint interval} \\ &= \frac{390}{60} = 6 \text{ checkpoints} \end{aligned}$$

• Data to be retransmitted

• Last checkpoint = 360MB

• Failure = 390MB

$$= 390 - 360$$

$$\Rightarrow 30\text{MB}$$

Importance of synchronization in the session layer

- Saves the communication process using checkpoints.
- Resume data transfer from the last checkpoint after failure
- Avoids retransmitting the entire file.
- Saves time and network bandwidth
- Improves communication reliability and efficiency

The session layer improves communication by using synchronization checkpoints. If a failure occurs, data transfer resumes from the last checkpoint, reducing retransmission time and improving network efficiency.

6. Given

→ Original file size = 900 MB

→ Compression = 40%.

→ Encryption overhead = 5% of compressed size

Presentation Layer (layer 6) is responsible for data formatting, compression, encryption, and decryption. It reduces the size of data before transmission and encrypts the data to provide security during communication.

Calculating the compressed file size

Compression = 40%.

$$= 900 - (40\% \text{ of } 900)$$

$$= 900 - (0.40 \times 900)$$

$$= 900 - 360$$

$$\therefore = 540 \text{ MB}$$

Calculating encryption overhead

Encryption overhead = 5% of 540 MB

$$= 0.05 \times 540$$

$$= 27 \text{ MB}$$

Calculating original file size

Original file size

$$= 900 \times 1024$$

$$= 921600$$

Calculating total percentage reduction

Reduction

$$= 900 - 567$$

$$= 333 \text{ MB}$$

Percentage reduction

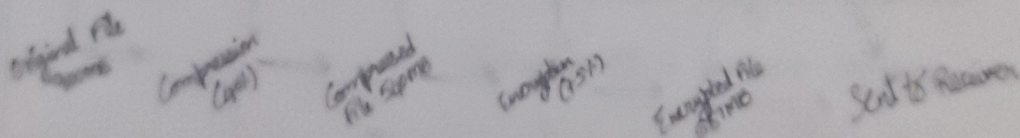
$$= (333 \div 900) \times 100$$

$$= 37\%$$

How the presentation layer improves transmission efficiency and security:

- Compresses data to reduce file size.
- Saves bandwidth and transmission time.
- Encrypts data at the receiver.
- Converts data into a standard format for communications.

The presentation layer improves communication by compressing data to reduce transmission time and encrypting data to ensure confidentiality and security during transmission.



7 Given :

- Data transferred = 3GB
- Throughput = 120Mbps
- Request size = 3MB

The application layer is the topmost layer of the OSI model. It provides network services directly to users and applications such as FTP, HTTP, SMTP, and DNS. It allows users to access files, email, and web services over a network.

To find ;

Calculations (No. of requests) & transmission time:

Let,

$$1\text{GB} = 1000\text{MB}$$

$$\therefore 3\text{GB} = 3000\text{MB}$$

$$3\text{GB} = 3072\text{MB} \approx 3000\text{MB}$$

$$\text{No. of requests} = \frac{\text{Total Data}}{\text{Request size}}$$

$$= 3000 \div 3$$

$$= 1000 \text{ requests}$$

Let,

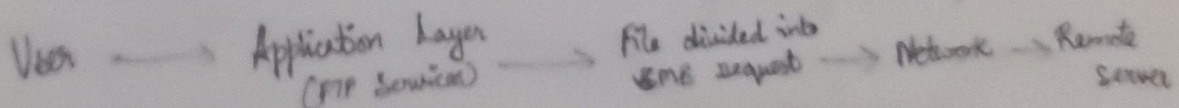
$$3000\text{MB} \times 8$$

$$= 24000\text{Mb}$$

$$\text{Transmission time} = \frac{\text{Total Data (Mb)}}{\text{Throughput (Mbps)}}$$

$$= 24000 / 120$$

$$= 200 \text{ seconds}$$



How the application layer provides reliable service

- Provides services like FTP, HTTP, SMTP, and DNS
- Allows users to access network resources
- Ensures reliable file transfer using FTP
- Supports user authentication and source reporting
- Provides an interface between users and the network.

The application layer provides network services data directly to users. It supports reliable file transfer, user authentication, and communication protocols such as FTP, ensuring efficient and reliable data transmission.

Given:

- Application Layer delay = 4ms
- Transport Layer delay = 3ms
- Network Layer delay = 5ms
- Data Link Layer delay = 2ms
- Physical Layer delay = 1ms
- Propagation delay = 18ms
- Transmission delay = 12ms

End to End Delay, is the total time taken for a packet to travel from the sender to the receiver. It includes the processing delay at different OSI layers, transmission delay, and propagation delay.

To find:

• Total processing Delay:

Processing Delay

$$4 + 3 + 5 + 2 + 1$$

$$= 15\text{ms}$$

Add Transmission Delay

$$15 + 12$$

$$= 27\text{ms}$$

End-to-End Delay

27 + 18

= 45 ms

Total End-to-End Delay = 45ms

How each layer contributes

- Application Layer
Provides network services to users.
- Transport Layer
Ensures reliable data transfer and flow control.
- Network Layer
Selects the best path and routes packets.
- Data Link layer
Converts packets into frames and detects errors.
- Physical Layer
Transmits data as electrical or optical signals.
- Transmission Delay
Time taken to place all bits onto the transmission medium.
- Propagation Delay
Time taken for the signal to travel from sender to receiver.

The total end-to-end delay is 45ms. Each OSI layer performs a specific function in communication, while transmission and propagation delays determine the time required for data to travel through the network.

Sender	Receiver	Delay	Total end-to-end
Application	Layer	= 4ms	
Transport		= 3ms	
Network		= 5ms	
Data		= 2ms	
Physical		= 1ms	
Transmission	Delay	= 20ms	
Propagation		= 2ms	

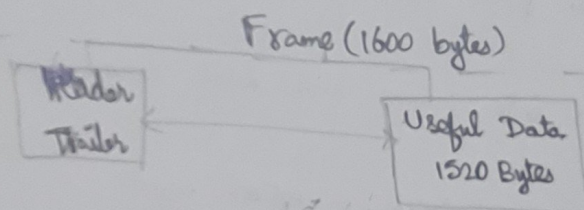
9. Given

- Useful data = 80MB
- Frame size = 1600 bytes
- Overhead per frame = 80 bytes

From the given data, to calculate the total number of frames, the total overhead transmitted, and the transmission efficiency

Protocol overhead

The extra information (headers & trailers) added to frame by different OSI layers for addressing, error detection, control. Although it increases the total data transmitted, it helps ensure reliable communication.



Total overhead = 80

bytes

Wk

$$\begin{aligned} 80\text{MB} &= 80 \times 1,000,000 \\ &= 80,000,000 \text{ bytes} \end{aligned}$$

$$\begin{aligned} \rightarrow \text{Useful data per frame} &= \text{Frame Size} - \text{Overhead} \\ &= 1600 - 80 \\ &\therefore 1520 \text{ bytes} \end{aligned}$$

$$\begin{aligned} \rightarrow \text{No. of frames} &= \text{Total Useful Data} \div \text{Useful Data per frame} \\ &= 80,000,000 \div 1520 \\ &\approx 52,632 \text{ frames} \end{aligned}$$

$$\begin{aligned} \rightarrow \text{Total Overhead} &= \text{No. of frame} \times \text{Overhead per frame} \\ &= 52,632 \times 80 \\ &= 4,210,560 \text{ bytes} \\ &= 4.21 \text{ MB} \end{aligned}$$

= 27MB

Multiple OSI Layers Working Together

- During data transmission, different OSI layers perform different functions.
- Presentation layer compresses and encrypts data.
- Transport Layer divides data into segments.
- Data Link Layer converts segments into frames and adds header/trailers.
- Physical Layer transmits the data as signals through the network.

To find the following,

• Compression = 30%

Compression offer size,

$$= 240 - (30\% \text{ of } 240)$$

$$= 240 - (0.30 \times 240)$$

$$= 240 - 72$$

$\therefore 168\text{MB}$ is the compressed size

$$168\text{MB} = 168 \times 1,000,000$$

$$\Rightarrow 168,000,000 \text{ bytes}$$

• Number of Segments = Compressed / Segments Size

$$= 168,000,000 / 1400$$

$$= 120,000 \text{ segments is the total.}$$

• Overhead per frame = 60 bytes

Total overhead

$$= 120,000 \times 60$$

$$= 7,200,000 \text{ bytes}$$

$$\Rightarrow \text{Total protocol overhead is } 7.2\text{MB}$$

Final data = Compressed data + overhead

$$= 168\text{MB} + 7.2\text{MB}$$

$$= 175.2\text{MB}$$

Calculate Transmission Efficiency

$$\begin{aligned}\text{Efficiency} &= (\text{Useful Data} \div \text{Total Data Transmitted}) \times 100 \\ &= 80,000,000 \div 84,210,560 \\ &= 95\% \end{aligned}$$

Efficiency

$$\begin{aligned}&= (80,000,000 \div 84,210,560) \times 100 \\ &= 95\% \text{ the total data transmitted}\end{aligned}$$

How protocol overhead affects network performance

- 1) Increases the total amount of data transmitted.
- 2) Uses extra network bandwidth.
- 3) Slightly increases transmission time.
- 4) Provides addressing and error detection
- 5) Ensures reliable and secure communication.

Explanation

Protocol overhead adds control information such as headers and trailers to each frame. Although it increases the amount of transmitted data, it improves communication by providing error detection, addressing, and reliable data delivery.

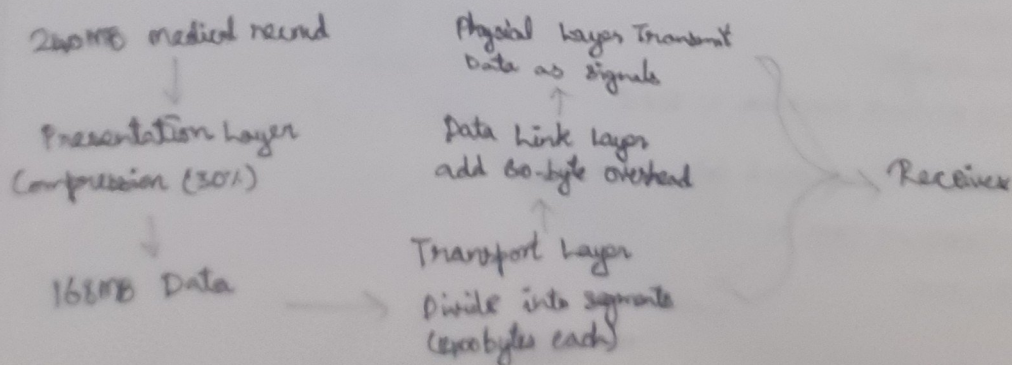
10. Given:

Medical record size = 240MB

Compression = 30%

Segment size = 1400bytes

Data link layer overhead = 60bytes per frame



How Multiple OSI Layers Work Together

- 1) Presentation layer compresses the data and improves security through encryption.
- 2) Transport layer divides the data into segments and ensures reliable delivery.
- 3) Data Link layer creates frames and adds headers/trailers for error detection.
- 4) Physical layer transmits the data as electrical, optical, or wireless signals.
- 5) All layers work together to provide secure, reliable, and efficient communication.

Conclusion

The OSI model ensures secure and reliable communication by allowing each layer to perform a specific task. The presentation layer compresses data, the transport layer provides reliable segmentation, the data link layer adds error detection, and the physical layer transmits the data through the network.

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